AGRI-GRID

Smart Agriculture & Food Security Ecosystem

Distributed Systems | IoT Telemetry | Predictive Analytics

Document Type	Software Engineering Report
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Roll Number	1512
Team Size	7 Members
Date Submitted	September 2026

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01 Project Overview

Agri-Grid is a **distributed, networked, and data-intensive** web application designed to modernize the agricultural supply chain. By replacing fragmented manual processes with an integrated digital ecosystem, Agri-Grid connects local farming operations directly with B2B markets, logistics networks, and remote agricultural experts.

The system is engineered to provide **end-to-end traceability**, ensuring food safety and provenance while optimizing resource allocation through predictive analytics.

> *"Agri-Grid replaces fragmented, opaque agricultural supply chains with a single, transparent, and intelligent digital ecosystem — from seed to table."*
>
> ***— Project Vision Statement***

Key Domain Tags

Precision Agriculture | B2B Marketplace | Cold-Chain Monitoring | Crop Diagnostics | RBAC Security | Escrow Payments | QR Provenance | Supply Chain

6	E2E	AI
User Roles	Traceability	Yield Prediction

02 The Problem Statement

The traditional agricultural supply chain suffers from severe bottlenecks that reduce farmer profitability, compromise food safety, and limit consumer transparency.

📊 Information Asymmetry

Farmers lack direct access to wholesale buyers, leading to reliance on middlemen and reduced profitability margins.

🚛 Logistical Inefficiencies

Without real-time transit tracking, perishable goods suffer from cold-chain failures and costly spoilage.

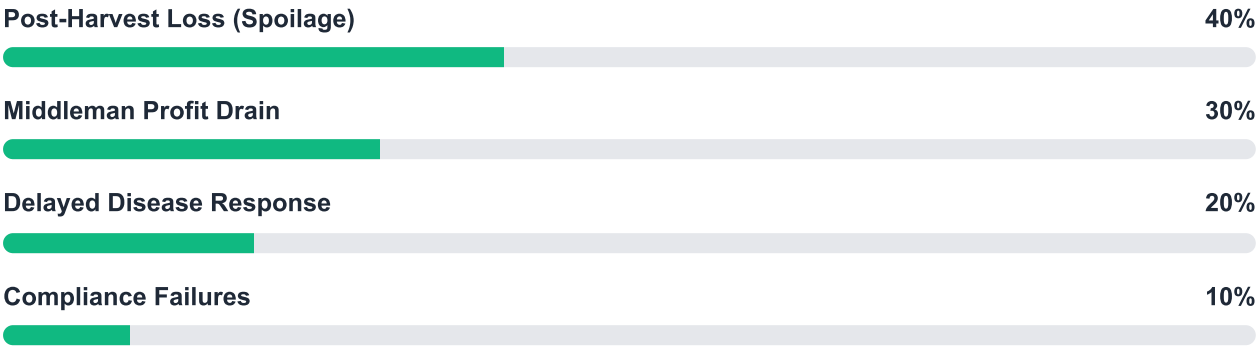
🔬 Delayed Diagnostics

Crop diseases spread rapidly when farmers cannot easily access remote agronomists for timely interventions.

🔍 Opaque Provenance

Consumers and compliance officers have no reliable way to verify origin, chemical usage, and food safety journey.

📉 Estimated Annual Impact of Supply Chain Inefficiencies



03 Software Engineering Methodology

To cope with the inevitable changing requirements of a large-scale system, this project rejects the traditional plan-driven Waterfall model in favor of modern iterative frameworks.

🔄 Incremental Model

System divided into manageable modules, each passing through requirements → design → implementation → testing cycles. Yields working software early.

⚡ Agile Integration

Iterative planning with continuous value delivery. Process adapts to reflect changing user feedback and evolving requirements at sprint boundaries.

✅ V-Model Testing

Testing planned in parallel with development phases for early defect detection. Continuous Verification & Validation (V&V) ensures quality.

04 System Roles & Core Features

The platform relies on strict **Role-Based Access Control (RBAC)** to ensure that users only interact with data relevant to their operational responsibilities.

The Farmer (10 Features)

Yield prediction dashboard, B2B marketplace listing tool, direct tele-consultation inbox, and environmental subsidy portal.

The B2B Buyer (8 Features)

Advanced produce catalog filtering, automated escrow contracting, and live cold-chain transit monitoring dashboard.

The Transporter (7 Features)

Mobile dispatch manifest, digital pickup/handshake verification, and IoT-simulated temperature telemetry logging.

The Agronomist (6 Features)

Remote triage queue, high-resolution diagnostic studio, and digital prescription pad for treatment plans.

Compliance Officer (5 Features)

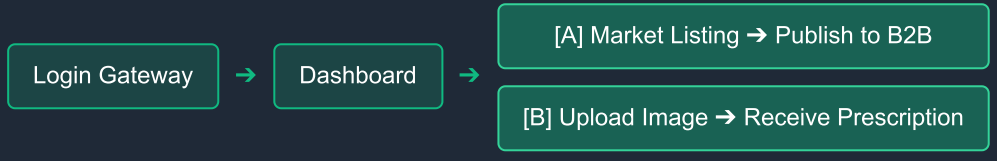
Subsidy audit board, environmental compliance verification, and transparent financial ledger exports.

Public Consumer (3 Features)

Unauthenticated QR-code batch lookup and visual farm-to-table provenance storyboards for transparency.

05 Visual User Journey Maps

Flow 1: The Farmer — Listing & Diagnostics



06 Technical Architecture & Toolchain

The ecosystem applies engineering discipline to construct a **secure, maintainable, and scalable** product. The development stack adheres to authorized tools and platforms.

📐 Modeling

StarUML / Draw.io

UML Use Case, Sequence, and Class diagrams to blueprint architecture.

📋 Project Mgmt

Trello / Jira

Agile sprint boards, task tracking, backlog management, and ownership.

🔀 Version Control

GitHub / GitLab

Source code governance with PRs, branching strategies, and issues.

⚙️ Backend & Data

Python / SQL

Predictive models, data pipelines, and highly structured relational database.

🧪 Testing

JUnit / Selenium

Backend logic verification and frontend UI validation for defect discovery.

📝 Documentation

Overleaf / Word

Collaborative SRS drafting and formal requirement specifications.

🏗️ Layered Architecture

PRESENTATION

Web Dashboard | Mobile | QR Scanner

▼

API GATEWAY

REST Endpoints | Auth | RBAC

▼

BUSINESS LOGIC

Yield Engine | Escrow | Diagnostics

▼

DATA & IoT

SQL Database | IoT Telemetry | Logs

📅 Roadmap (Sprints)

SPRINT 1-2

Foundation & Auth

Database schema, RBAC engine, login/registration, base skeleton.

SPRINT 3-4

Farmer & Market

Crop listing CRUD, yield prediction, B2B catalog search engine.

SPRINT 5-6

Logistics & IoT

Transporter dispatch, GPS sim, telemetry, delivery handshake.

SPRINT 7-8

Diagnostics & Audit

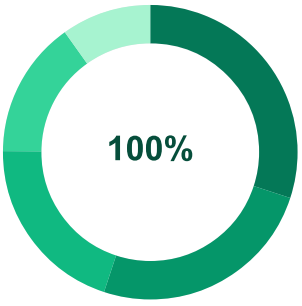
Image-based crop diagnosis, agronomist queue, QR provenance.

07 Team Overview & Summary

👥 Project Team Roster

AM	**Ameer Muawiyah** Project Lead · Roll No: 1512	
S1	**Student 1** Team Member	
S2	**Student 2** Team Member	
S3	**Student 3** Team Member	
S4	**Student 4** Team Member	
S5	**Student 5** Team Member	
S6	**Student 6** Team Member	

📊 Module Effort Distribution



📋 Allocation Legend

- **Marketplace Module** — 30%
- **Logistics & IoT** — 25%
- **Crop Diagnostics** — 20%
- **Auth & RBAC** — 15%
- **Compliance & Audit** — 10%

🌿

Agri-Grid

From Seed to Table — Transparently